THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $(5,\ 8)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
2	**(b)** lies on the line	RHS at $x = 7$ is 15, equal to 15
3	**(d)** 4	number of places $= \max(a, b)$ in $2^a 5^b$
4	**(d)** $\frac{1}{6}$	standard method ⇒ $\frac{1}{6}$
5	**(a)** $(x+y)^2 = x^2 + 2xy + y^2$	true for all values of the variables
6	**(a)** 75°	angle at the centre is double the angle on the circle
7	**(b)** 60 cm^2	$s = 20$, area $= \sqrt{20 \cdot 12 \cdot 5 \cdot 3} = 60$
8	**(b)** $\frac{5}{6}$	favourable outcomes: 1, 2, 3, 4, 6 ⇒ 5 out of 6
9	**(a)** neither an AP nor a GP	differences 2,3,4,5 are not constant
10	**(d)** composite	$2^{32} + 1 = 641 \times 6700417$
11	**(d)** 76.4	$\frac{15 \times 82 + 35 \times 74}{15 + 35} = 76.4$
12	**(b)** 14%	$\frac{250 \times 20 + 750 \times 12}{250 + 750} = 14\%$
13	**(d)** 12	repeated reduction ends at gcd(12, 0)
14	**(a)** 75	the rightmost element is the gcd
15	**(b)** 5	the chain of reductions ends at gcd(3, 0)
16	**(d)** parallelogram	Theorem 4 (converse of (c))
17	**(c)** intersect each other	test for convexity
18	**(c)** A is true but R is false	A is true (coincident lines); R is false because parallel lines give none
19	**(d)** no solution	compare a1/a2, b1/b2, c1/c2
20	**(b)** 308 cm^3	$\pi r^2 h = 308$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$PA^2 = 1 + 9 = 10$; $PB^2 = 9 + 1 = 10$. Yes, P is equidistant.	1+1
22	Slope $= -4$, y-intercept $= 9$; negative slope ⇒ linear decay.	1+1
23	$\angle BCD = 180^\circ - 70^\circ = 110^\circ$; opposite angle $\angle A = 180^\circ - 110^\circ = 70^\circ$.	1+1
24	Stage n has 3^n black triangles: Stage 4 has 81 and Stage 5 has 243.	1+1
25	Converse: If a number is divisible by both 2 and 3, then it is divisible by 6 (1). Both are true because 2 and 3 have no common factor (1).	1+1
26	$\frac{5\times4+4\times3+2\times2}{9} = \frac{36}{9} = \mathbf{4}$ (2).	1+1
27	Ratios: $\frac{2}{4} = \frac{3}{6} = \frac{9}{18} = \frac{1}{2}$ (1). The lines coincide, so there are infinitely many solutions (1).	1+1
28	Surface area becomes 4 times (1); volume becomes 8 times (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) gcd(144, 108) (1). (ii) $144 = 108 \times 1 + 36$; $108 = 36 \times 3 + 0$ ⇒ gcd $= 36$ (1). (iii) Apples $144/36 = 4$ boxes, oranges $108/36 = 3$ boxes (1).	1+1+1
30	(i) $70 + 80 + 100 + 110 = 360^\circ$ (1). (ii) 360° — a full turn, no gap or overlap (1). (iii) Yes, the arrangement can be repeated (rotate a copy by 180° about the midpoint of a side) so any quadrilateral tiles the plane (1).	1+1+1
31	$y = 3, 5, 11, 15, 19, 25, 41$. $(4, 9)$: $2(4) + 1 = 9$ ✓ lies on the line; $(6, 14)$: $2(6) + 1 = 13 \neq 14$ ×.	2+1
32	$= pqx^2 + (pb + qa)x + ab$. For $(2x + 3)(3x + 1)$: $6x^2 + (2 + 9)x + 3 = 6x^2 + 11x + 3$.	1+2

33	$RB = \sqrt{9+16} = 5$; $BC = \sqrt{36+49} = \sqrt{85}$; $RC = \sqrt{81+121} = \sqrt{202}$. $(RB+BC)^2 = 25 + 85 + 10\sqrt{85} = 110 + 10\sqrt{85} \neq 202 = RC^2$, so $RB + BC \neq RC$. Hence the points are not collinear.	1+1+1
34	Suppose they meet at a point: then $mx + d_1 = mx + d_2$ (1). This gives $d_1 = d_2$ (1), which contradicts $d_1 \neq d_2$; so they have no common point and are parallel (1).	1+1+1
35	If $5 - \sqrt{3} = r$ (rational) then $\sqrt{3} = 5 - r$ is rational — contradiction. Hence irrational.	1+1+1
	OR: LHS $= \frac{1}{2} \times \frac{7}{12} = \frac{7}{24}$; RHS $= \frac{1}{8} + \frac{1}{6} = \frac{3}{24} + \frac{4}{24} = \frac{7}{24}$ ✓	1+1+1
36	Slope between first two $= 3$; between the next two $= 6$ (2). Unequal slopes ⇒ not collinear (1).	2+1
	OR: $t_n + t_{n-1} = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} = n^2$ (2). $55 + 45 = 100 = 10^2$ ✓ (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{46x + 36(50-x)}{50} = 40$ (1). (ii) $10x = 200 \Rightarrow x = 20$ males; 30 females (2). (iii) $\frac{30 \times 36 + 42}{31} = \frac{1122}{31} \approx 36.19$ kg (2).	1+2+2
	OR: (i) $\frac{100 \times 0.4 + 150 \times 0.2 + 0.6y}{250+y} = 0.4$ (1½). (ii) $70 + 0.6y = 0.4(250 + y) = 100 + 0.4y \Rightarrow 0.2y = 30 \Rightarrow y = 150$ kg (2). (iii) Silver $= 40 + 30 + 90 = 160$ kg; total $= 400$ kg; $\frac{160}{400} = 0.4$ ✓ (1½).	1½+2+1½
38	(i) gcd(84, 126) $= 42$ (1). (ii) Common divisors $= [1, 2, 3, 6, 7, 14, 21, 42]$ (1). (iii) Boys $84/42 = 2$ rows; girls $126/42 = 3$ rows (1). (iv) The division algorithm — its number of steps depends on the digits, not the values, so it stays fast (2).	1+1+1+2
	OR: (i) $k = 2$ ✓, $k = 5$ ✓, $k = 10$ ✓ (others up to 20 ×) (1). (ii) Final value $10 = \gcd(20, 30)$ (1). (iii) $k = 2, \ldots, 20$: 19 values (1). (iv) Use $\gcd(m, n) = \gcd(n, m \bmod n)$: $30 = 20 \times 1 + 10$; $20 = 10 \times 2 + 0$ gives 10 in 2 steps (2).	1+1+1+2
39	(i) $F = (6, 4)$. (ii) $FL_1 = 6 = FL_2$ ✓. (iii) Area $= 96$ sq. units; $OB = \sqrt{144+64} = \sqrt{208} = 4\sqrt{13}$ units.	1+2+2
	OR: $AB = 4$, $BC = \sqrt{4+4} = 2\sqrt{2}$, $CA = 2\sqrt{2}$; $PQ = 8$, $QR = \sqrt{16+16} = 4\sqrt{2}$, $RP = 4\sqrt{2}$. Ratios $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} = \frac{1}{2}$ ⇒ similar.	3+2
40	(i) No, π is irrational. (ii) $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} = \frac{105-35+21-15}{105} = \frac{76}{105}$; times $4 = \frac{304}{105} \approx 2.895$ — a rough value, improving with more terms. (iii) Lambert, 1761.	1+2+2
	OR: (i) For $a < b$ the average $\frac{a+b}{2}$ is rational and lies between them (3). (ii) $\frac{1}{2}\left(-\frac{3}{12} + \frac{5}{12}\right) = \frac{1}{12}$; it lies to the right of 0, one-twelfth of a unit from the origin (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.